AI Hackathon Grand Finale

Official Brief

Challenge: The Adaptive Malaysia

Challenge Overview

The Adaptive Malaysia calls on builders to reimagine how systems respond to real-world complexity — from urban mobility to public services and business operations.

In today's environment, many systems remain reactive, fragmented, and dependent on manual coordination. As conditions change, responses are often delayed, inefficient, or disconnected across teams and platforms.

This challenge invites you to design Agentic AI systems that can:

- Sense changing conditions
- Interpret data in context
- Make decisions
- Trigger coordinated actions in real time

Your goal is to explore how intelligent coordination can make Malaysian systems more responsive, connected, and efficient.

What is an Agentic AI System?

An agentic AI system is one that perceives inputs from its environment, reasons about them, makes decisions, and takes actions — without requiring a human to manually trigger each step.

It is not a chatbot that waits for instructions. It is not a dashboard that displays information. It is a system that thinks, decides, and acts.

A strong agentic system will complete at least three distinct decision or action steps autonomously — for example: detecting a condition, evaluating options, and triggering a response.

Exploration Areas

You may explore problems within areas such as:

- Urban mobility and transport systems
- Public services and citizen-facing operations
- Enterprise workflows and business operations
- Digital systems requiring real-time coordination

These areas are intentionally broad. You are expected to identify a specific, high-impact scenario within your chosen area, and design a solution that addresses it with clarity and depth.

Example Scenario Hints (for inspiration only)

- A sudden disruption causes abnormal traffic congestion requiring real-time coordination
- A surge in service requests overwhelms an organisation, requiring prioritisation and intelligent routing
- An operational breakdown (staff shortage, system delay) requires dynamic reallocation of resources

These are examples only. You are encouraged to define your own scenario.

Your Task

- Choose ONE problem scenario
- Define it clearly
- Design an Agentic AI system that solves it

You are NOT solving an entire domain. You are solving one meaningful problem exceptionally well.

Core Expectations

1. Clear Problem Definition

- Specific scenario | Identified user or system | Well-scoped problem

2. End-to-End System Thinking

- Input → Processing → Decision → Action | Clear interaction between components

3. Agentic Behaviour (CRITICAL)

- Make decisions
- Execute multi-step workflows (minimum 3 distinct decision or action steps)
- Trigger at least one autonomous or system-driven action

4. Real-World Applicability

- Where this system fits in a real organisation
- How it would be used and what value it delivers

5. Scalability & Practicality

- Can this scale? | Is it feasible? | Is it cost-conscious?

Data & Environment

All datasets must be derived from open-source platforms such as Kaggle, GitHub, official government websites, or public APIs. Fully synthetic or randomly generated data is not allowed.

Your data must be realistic and contextually appropriate, reflecting the kind of information a real Malaysian organisation would actually work with.

You must also demonstrate how your system would function when integrated with live data sources.

Deployment & Hosting (Mandatory)

- The solution must be deployed on a live hosting platform (e.g. AWS, Azure, GCP, Vercel, Railway, or equivalent)
- **Localhost is NOT acceptable**
- The system must be fully functional and accessible during the Friday presentation
- It should resemble a working enterprise-grade solution integrated with live or open-source data
- *A recorded backup video is strongly recommended as contingency if live connectivity fails*

What We Provide

We want to set you up for success. Here is what is available to you:

- Online mentoring session: Tuesday, 6 May 2026 at 9:00 AM
- Each team will be reimbursed up to USD 20 for API key usage (e.g. Claude, Gemini, ChatGPT, or similar LLM services) towards building and running your solution
- A dedicated point of contact for technical queries during the hackathon (details to be shared)

Bring your questions about problem scope or system design to the mentoring session — that is what it is for.

What Will Be Penalised

- Generic or overly broad problems
- Static dashboards with no action layer
- Chatbot-only solutions without decision logic
- Systems that cannot clearly explain how they operate
- Overly complex ideas with weak execution
- Solutions not deployed or running only on localhost

- Use of fully synthetic or randomly generated data not sourced from open platforms

Submission Requirements — Thursday, 7 May 2026 at 7:00 PM (Strict Deadline)

Required:

- Final presentation slides (PDF or PPT)
- Clearly defined problem statement (within slides)
- System workflow and architecture (within slides)
- Live demo link — must be accessible at time of submission

Optional but Recommended:

- Short recorded demo video as backup

Full documentation, reports, and code submission are NOT required.

Final Deliverables — Pitch Day (Friday, 9 May 2026)

- A live functional demo (recorded backup accepted only if live fails)
- Clear explanation of the problem, solution, and impact
- System workflow and architecture
- Ability to answer Q&A; from judges

Evaluation Criteria — 100 Points Total

#	Criteria	Pts
1	Problem Relevance & Impact Importance, realism, clear stakeholder, real-world value	15
2	AI Agent Design & Intelligence True agentic behaviour, multi-step reasoning (min 3 steps), autonomous actions, tool/API use	20
3	Technical Execution System reliability, clear architecture, data/integration handling	15
4	Product Thinking & User Experience Usable, intuitive, logical workflow	15
5	Innovation & Differentiation Originality, creative problem-solving, beyond basics	10
6	Scalability & Real-World Feasibility Scalable, cost-aware, practical deployment	10

7	Responsible AI & Risk Awareness Incorrect outputs, data sensitivity, reliability	5
8	Demo & Presentation Compelling demo, real usage shown, easy to follow	10
	TOTAL	**100**

Important Guidance

- Solve ONE problem well
- Focus on system logic, not just visuals
- Ensure your solution is usable and intuitive
- Demonstrate how your system thinks, decides, and acts
- Keep your solution realistic and explainable

Timeline

Tuesday, 6 May 2026 — 9:00 AM	Online mentoring session begins
Thursday, 8 May 2026 — 7:00 PM	Submission deadline (strict)
Friday, 9 May 2026	Final presentation and judging

You are not being evaluated as students.

You are being evaluated as builders capable of designing systems for real-world environments.

We are not evaluating how much you build,

but how well your system thinks, decides, and operates under real conditions.